

Project

Algorithms & Data Structures Database Systems

- Group 36
- 3th BSc Business Engineering
- External Partner: Frédéric Pattyn



Presentation Overview



WHAT IS
REVIEWR



OUR TEAM



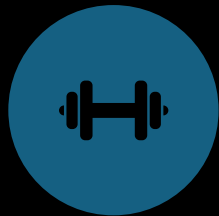
PROBLEM &
SOLUTION



PROJECT
METHODOLOGY



RESULTS



STRENGTHS &
LIMITATIONS



FUTURE

What Is ReviewR?



Connects Worlds: A platform that directly links researchers with companies.



AI Analysis: Our AI reads papers and scores them on **Business Value** and **Academic Quality**.



Easy Discovery: Helps companies find relevant new technologies fast.



Speed: Turns a complex PDF into a clear summary in minutes.

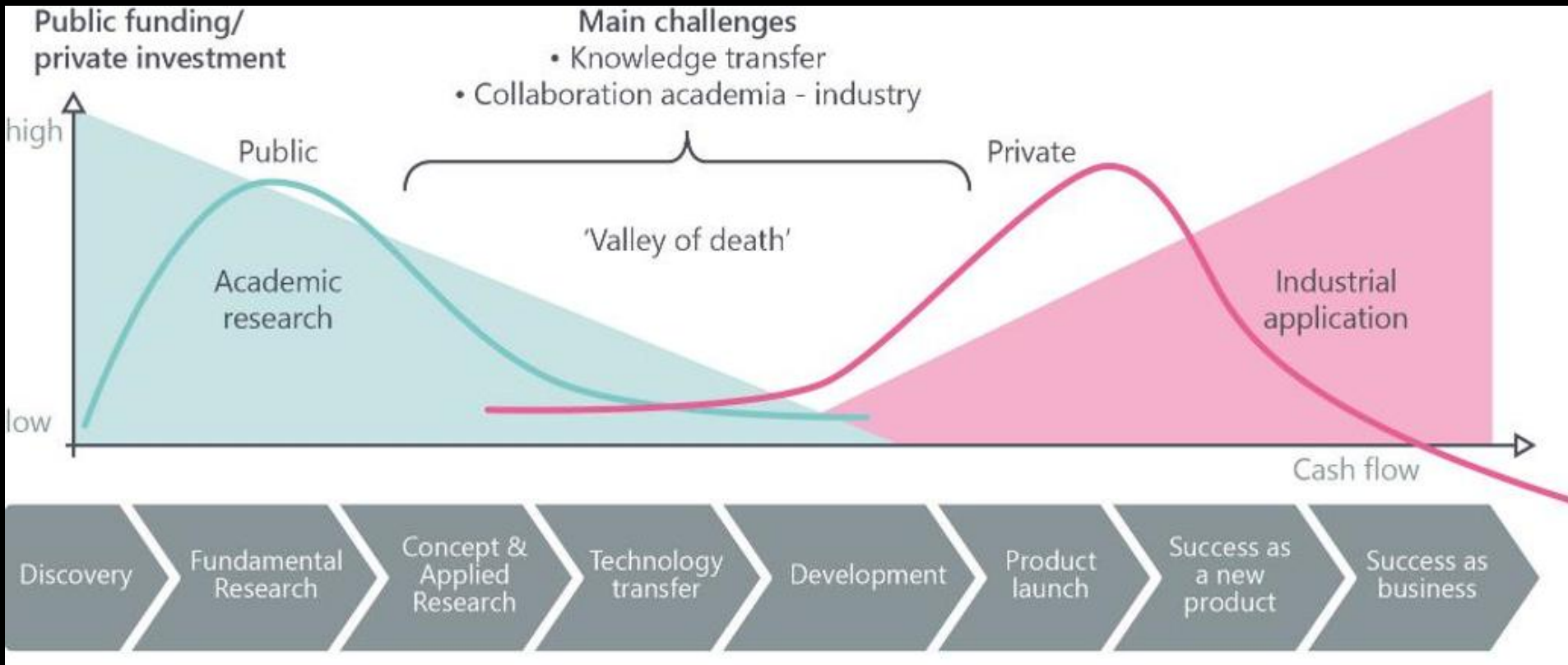
Our team

- **Manager** – Emiel Van hauwermeiren
- **Database Designer** – Gaetan Focquet
- **Programmers/ UI-designers** - Manu Embo, Mauro Deslee and Thomas Waelkens
- External partner - Frédéric Pattyn



The Problem

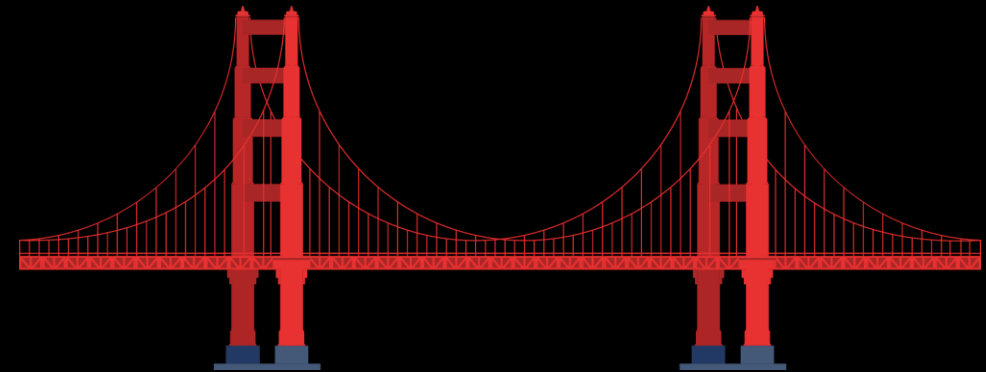
The innovation Gap



The Problem

Academic research fails to reach the industry!

- Hard to find
- Hard to use
- No clear match



Solution? -> REVIEWR

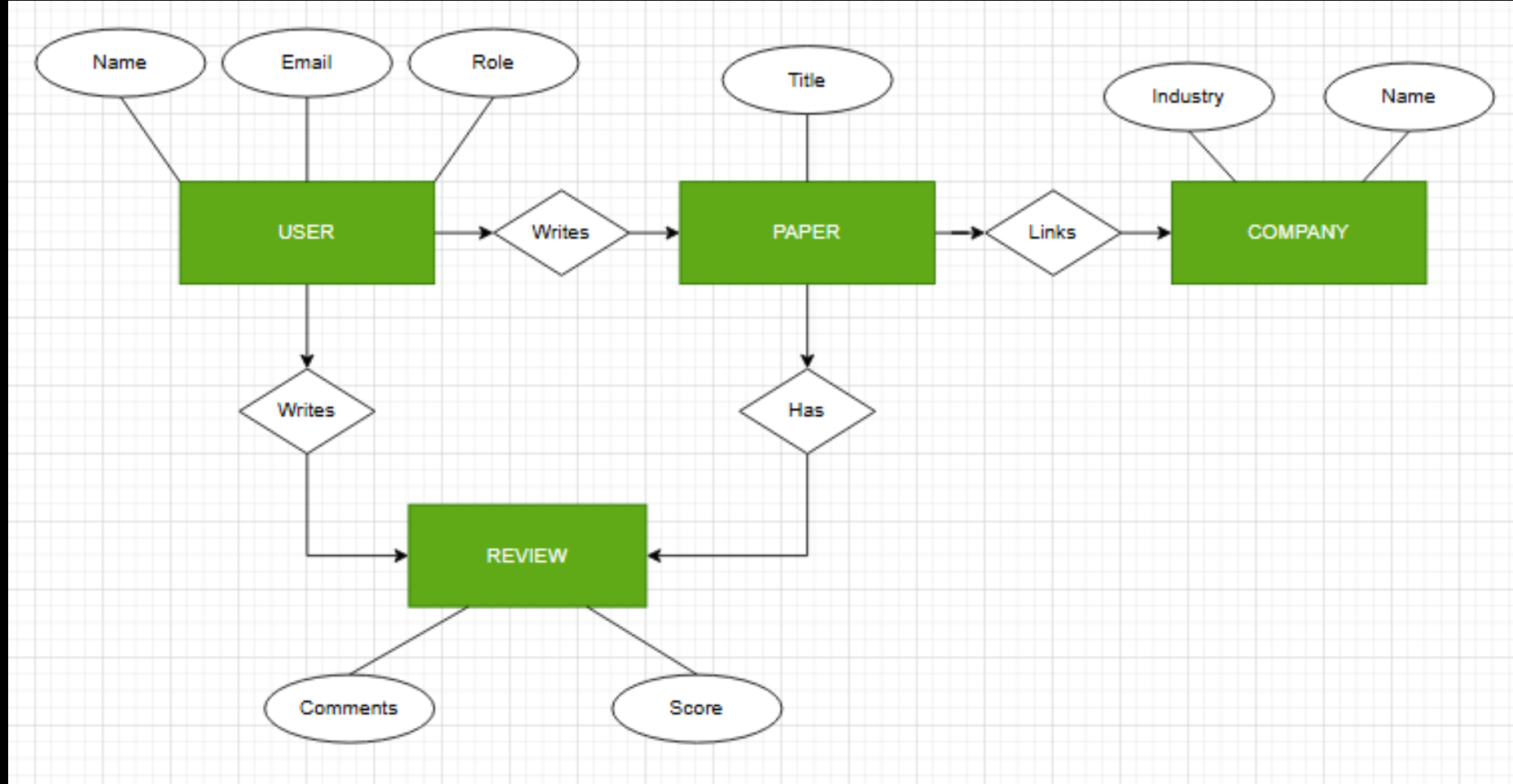
- Bridging research and real-world innovation
- Smart discovery
- AI-Powered summaries
- Trusted quality
- Open access

PROJECT METHODOLOGY

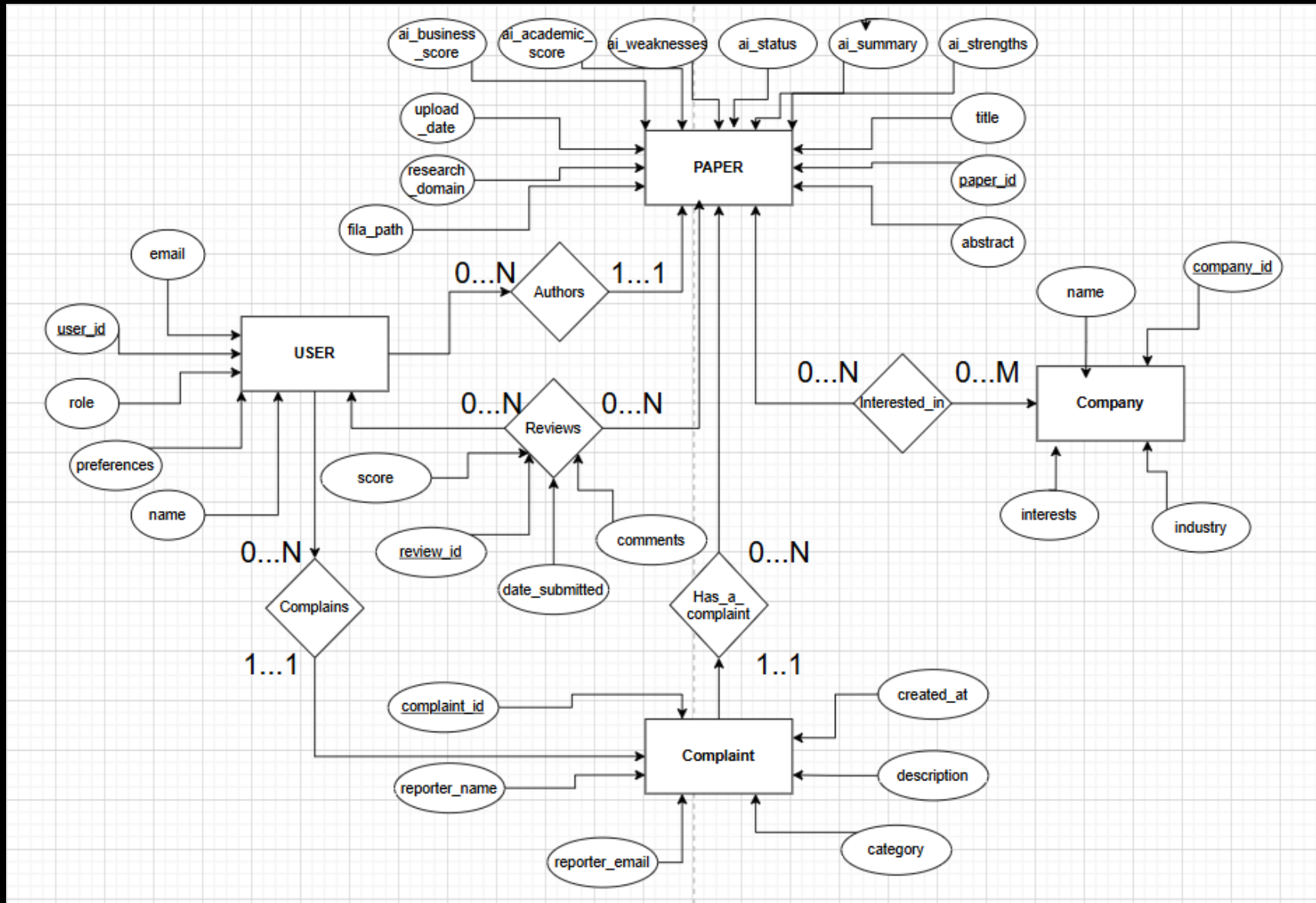
- 1) Construction ERD
- 2) UI example via lovable
- 3) System Architecture
- 4) AI Implementation
- 5) Our Algorithm



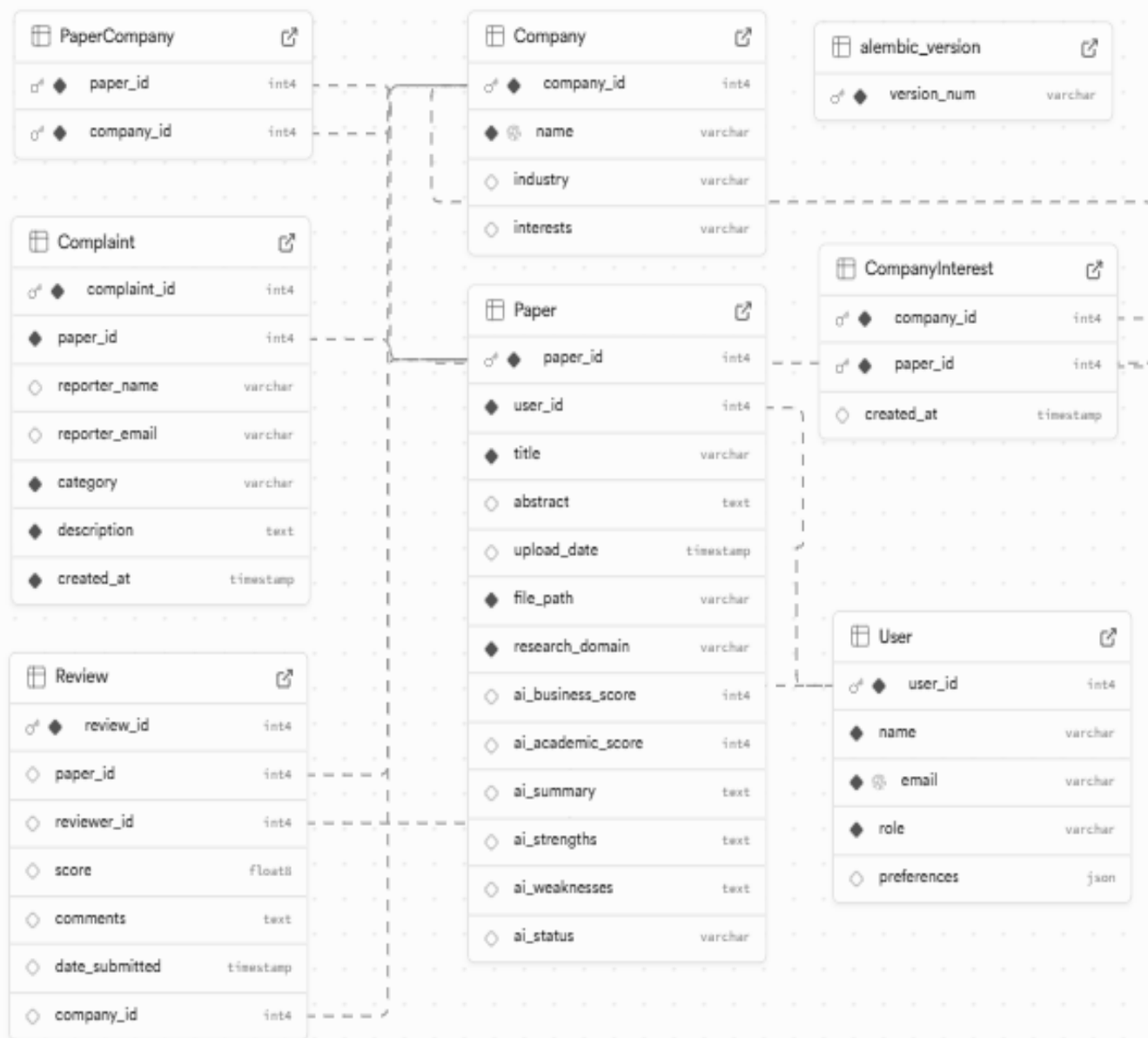
Conceptual ER (first version)



Conceptual ER (final version)



Supabase



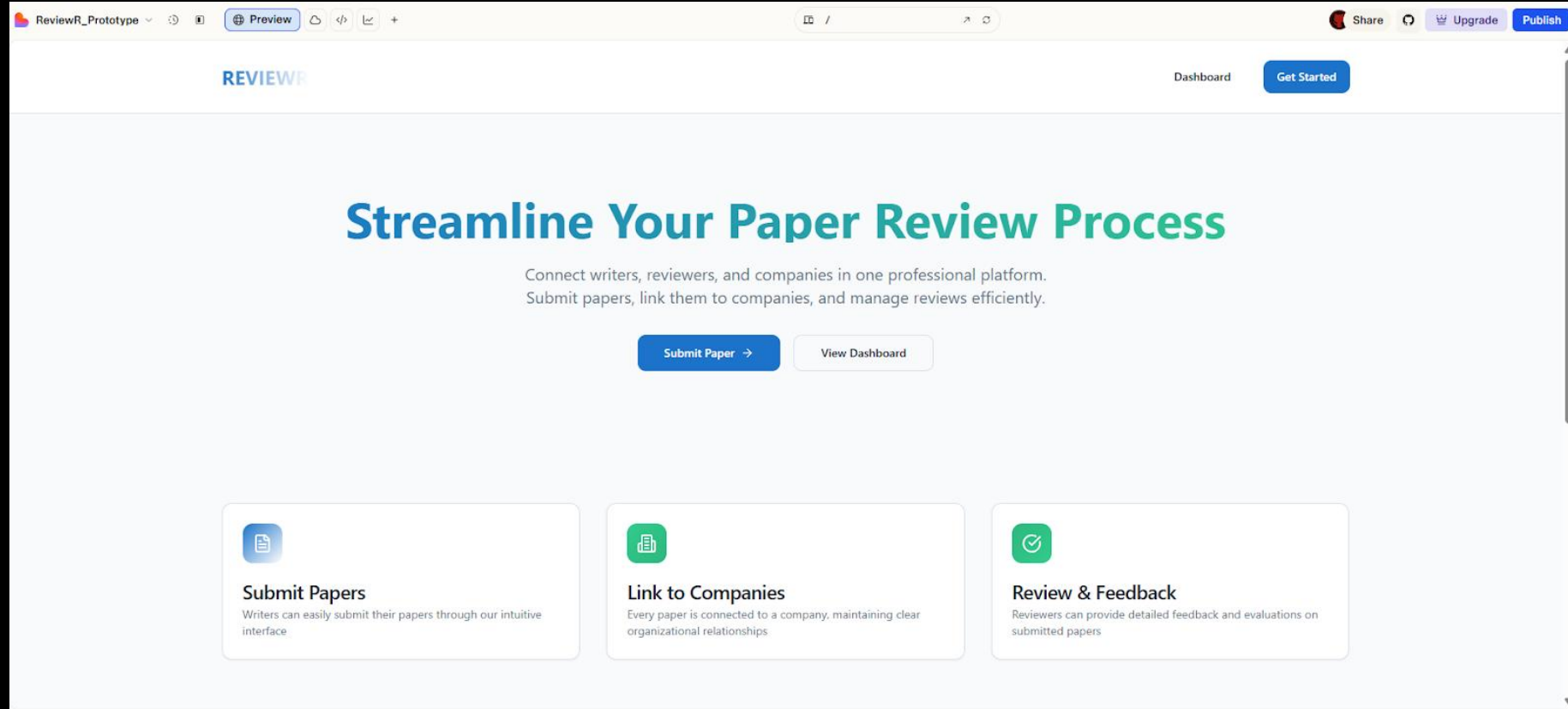
- PostgreSQL: relational database with foreign keys and constraints.
- SQLAlchemy ORM: converts Python to SQL

Why Supabase?

- PostgreSQL-based
 - reliability & scalability
- Built-in authentication & authorization
 - secure user management without custom backend code
- Real-time database support
 - live updating



2) UI example via lovable



3) System architecture

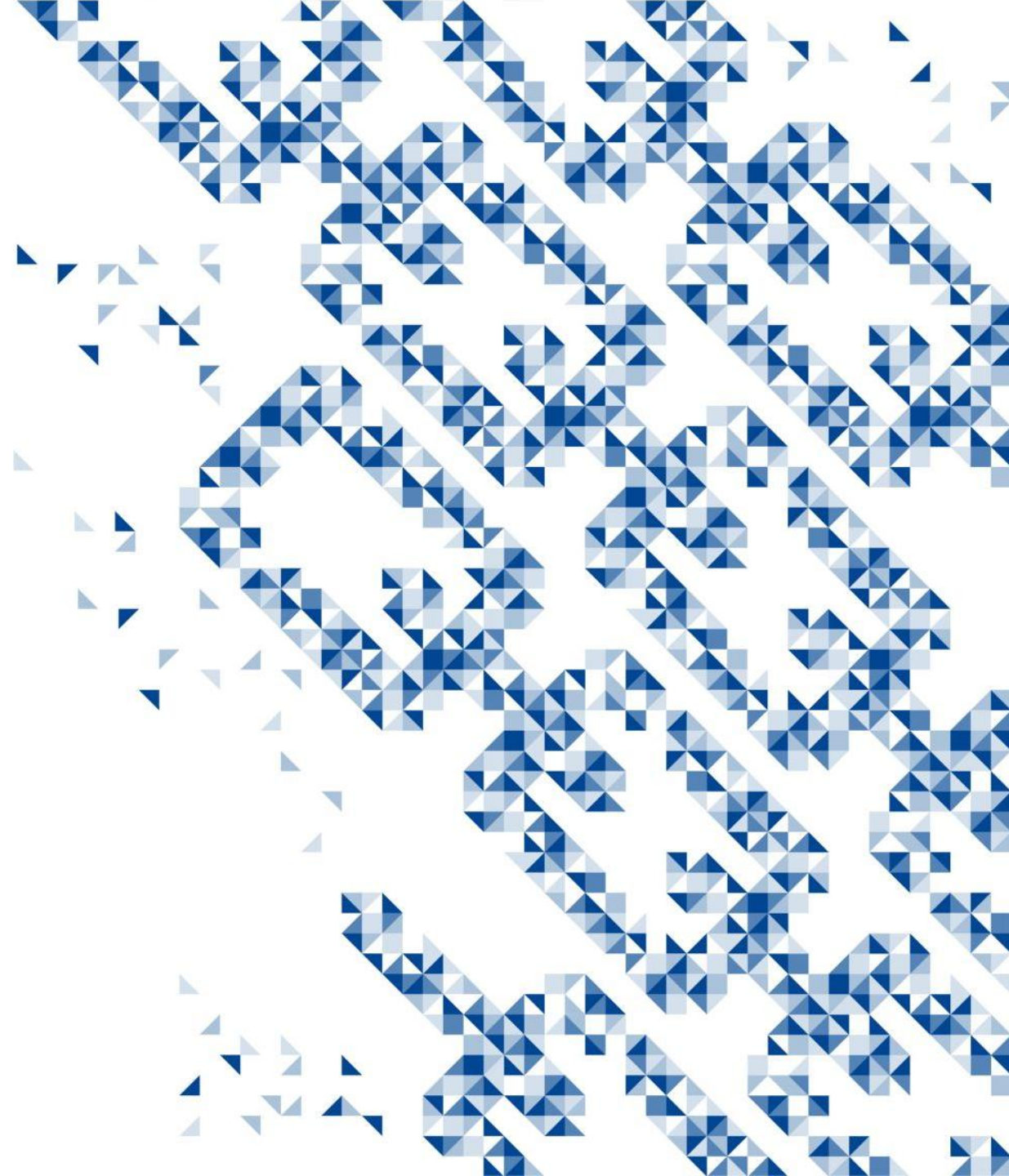
- **Presentation/interaction layer (frontend)**
 - Responsive & Interactive User Interface
 - HTML, CSS
- **Application layer (backend)**
 - Brain of the application
 - Python, Flask Framework.
- **Data layer**
 - Secure Cloud Storage & Relational Data
 - Supabase (PostgreSQL), Cloud Storage via buckets

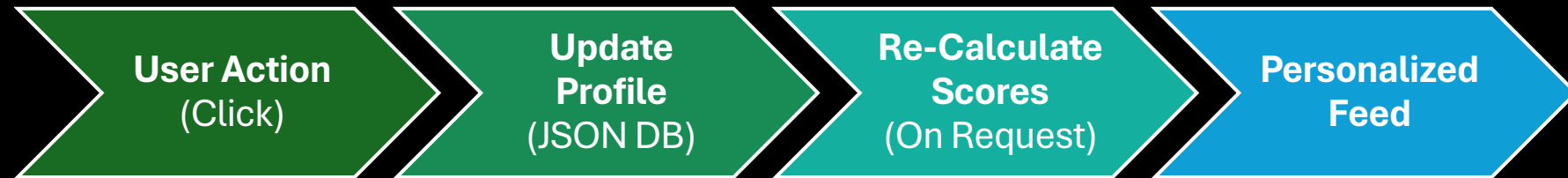
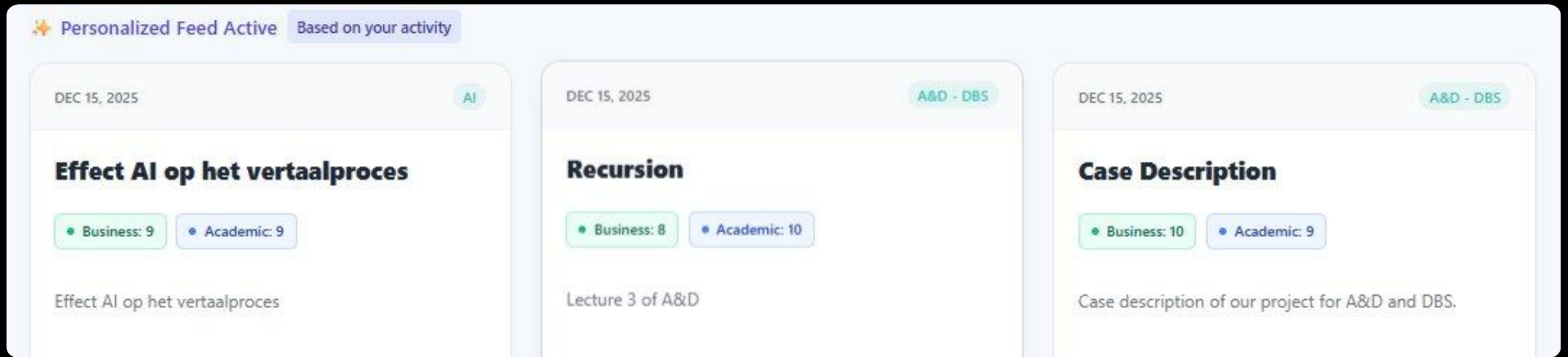


4) AI Implementation

Very important for our project

- 1) **Initiation:** User uploads PDF -> System secures filename & stores in Cloud.
- 2) **Extract:** Python reads the raw stream (in-memory) -> Extracts text.
- 3) **Analyze:** Text is sent to the LLM (gemini) with a specific prompt for "Business" vs "Academic" criteria.
- 4) **Display:** Scores are returned in JSON and rendered instantly on the Dashboard





5) Our Algorithm: A Personalized Feed

5) Our Algorithm (*Hybrid Recommendation*)

$$\text{Score} = (W_p \times \text{User Preference}) + (W_{pop} \times \text{Popularity}) + (W_r \times \text{Recency})$$

"A weighted multi-variable scoring model designed to balance personalization with trending content."

User Preference (50% Weight):

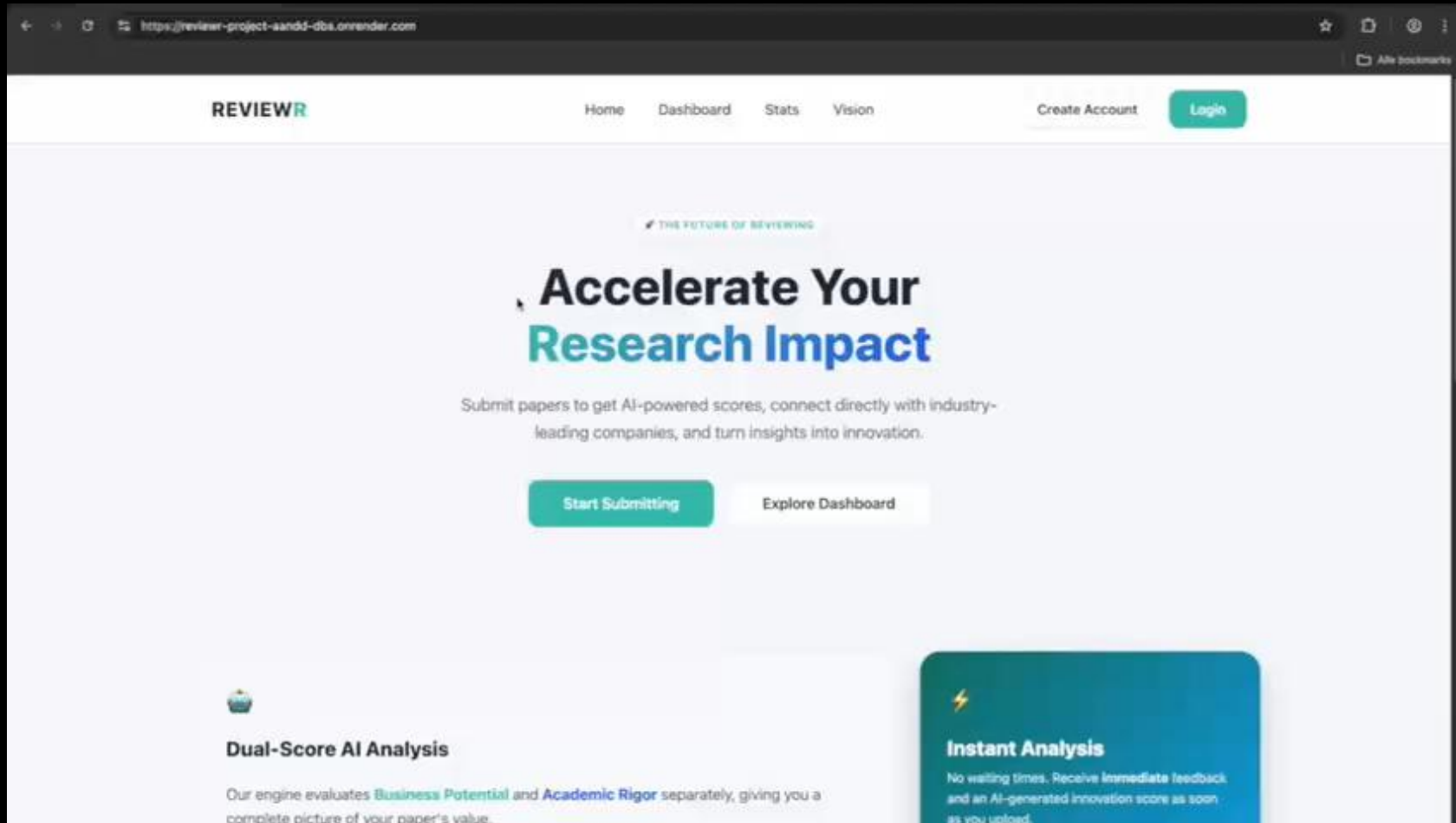
- **Input:** Click History
- **Logic:** Dynamic User Profile (JSON)
- **Example:** User clicks "AI"
 - "AI" domain score increases

Social Popularity (30% Weight):

- **Input:** Reviews & Rating
- **Logic:** Logarithmic Scale x Avg Rating
- **Goal:** Surface high-quality, peer-validated research.

Temporal Decay (20% Weight):

- **Input:** Upload Date vs. Current Date
- **Logic:** Linear decay function
- **Goal:** Prioritize cutting-edge research over outdated papers.



Result



Strengths

- Clear role-based workflows
 - User-friendly layout
 - Human and AI evaluation combined
 - AI score divided in academic and business score
 - Transparent feedback: visible reviews & scores
- 

Current Limitations

- Limited personalization and matchmaking
- No author/paper/company verification <-> report function
- No communication options between users
- No reputation or trust system
- Demo-level security

Scalability

1. System & Architecture

Client-side uploads



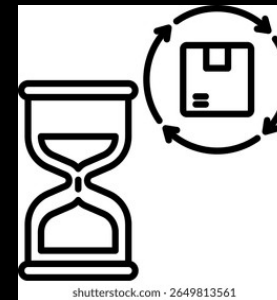
Offloading server



Asynchronous Processing



Put heavy AI-analysis in a queue



Scalability

2. Matchmaking:

CURRENTLY

-Recommended via 'RESEARCH DOMAIN'

`Paper.research_domain IN company.interests`

-Manual matching via 'INTERESTED' button

INTERESTED

POSSIBILITIES

-Company interest history

-More preferences than just "research domain"

-Behavior-based personalization

-More complex personalized recommendation algorithms (Netflix/Youtube-style)

Scalability

3. Personalization of accounts

Company:

- Industry & sub- industry
- Risk appetite
- Commercial goals
- Budget range
- ...

Researcher:

- Degree
- Field of study
- Academic recognitions
- Key publications
- ...

User:

- Expertise domains
- Review quality
- Interests
- ...

Better personalization results in better matching

Future enhancements

- Leveling system based on uploads and reviews
- Points & badges to reward active contributors
- Reputation scores to increase trust and quality



What did we learn during the project?

- Collaborative development using GitHub
- Building a backend with Supabase & PostgreSQL
- Writing and debugging code in Visual Studio Code
- Integrating AI into a real application
- Working as a full-stack team

Thanks for
listening!

REVIEWR

